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MILITARY SPECIFICATION

CLOTH, SILK (FOR AMMUNITION PARACHUTES)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope.—This specification covers silk cloth for making ammunition parachutes.

1.2 Classification.—The silk cloth shall be of the following classes, as specified (see 6.):

Class 1—high air-permeability cloth.
Class 2—low air-permeability cloth.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 —Boxes, Wood - Cleated-Fiberboard.
NN-B-601 —Boxes, Wood - Cleated-Plywood, For Domestic Shipment.
NN-B-621 ——Boxes, Wood, Nailed and Lock Corner.
UU-P-268 —Paper, Kraft, Wrapping.
UU-T-111 —Tape, Paper, Gummed (Kraft).
CCC-T-191 —Textile Test Methods.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The silk cloth shall be manufactured from filature silk.

3.2 Width.—The width of the cloth shall be as specified by the procuring agency (see 6.1), and shall be determined as specified in 4.4.1. The specified width shall have a tolerance of ± 0.5 inch.

3.3 Physical properties.—The cloth shall conform in its physical properties to the requirements specified in table I, when tested as specified in the applicable test paragraph (6.2).

3.4 Acidity or alkalinity.—The acidity or alkalinity of the cloth shall be not more than 0.05 percent, determined as specified in 4.4.5.

3.5 Workmanship.—The silk cloth shall be uniformly woven, shall contain no sizing materials, and shall be free of foreign inclusions.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—A lot shall consist of not more than 100,000 yards of silk cloth of the same class.

4.2 Sampling.—Ten percent of the rolls of cloth comprised in a lot shall be selected at

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TABLE I.—*Requirements for the physical properties of the silk cloth*

Property	Class 1		Class 2		Test paragraph
	Minimum	Maximum	Minimum	Maximum	
Weight, ounces per square yard	0.48	0.60	0.68	0.78	4.4.2
Breaking strength					
Warp, pounds	16	---	22	---	4.4.3
Filling, pounds	16	---	22	---	
Air permeability, cubic feet per minute per square foot.	340	550	175	330	4.4.4

random. However, not less than 50 rolls shall be selected from any lot. Approximately 1 yard of cloth shall be cut from each selected roll. Each of these primary samples shall be labeled so that the roll from which it was obtained can be identified.

4.2.1 Approximately one-half yard of cloth shall be removed from each of the primary samples, and a composite sample for the lot shall be made up of all the subsamples so obtained. The composite sample shall be placed in a securely closed container, with the container labeled to show the name of the material, class, manufacturer, lot number, contract or purchase order number, and number of yards in the lot.

4.2.2 All acceptance tests shall be performed on the composite sample, using as many of the individual subsamples as is practicable. The primary samples shall be held for possible future examination should the composite sample fail to meet the specified requirements.

4.3 **Inspection.**—Inspection shall be at the point of delivery unless otherwise specified.

4.3.1 *Inspection procedures.* — For Navy purchases the inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.4 **Tests.**

4.4.1 *Width.*—The width of the cloth shall be determined by method 5020 of Specification CCC-T-191.

4.4.2 *Weight.* — The weight of the cloth shall be determined by method 5041 (small specimen method) of Specification CCC-T-191.

4.4.3 *Breaking strength.* — The breaking strength of the cloth shall be determined by method 5100 (grab method) of Specification CCC-T-191.

4.4.4 *Air permeability.*—The air permeability of the cloth shall be determined by method 5450 of Specification CCC-T-191. Alternate method 5452 of Specification CCC-T-191 may be used, provided the results of the test are converted to cubic feet of air per minute per square foot of fabric (see 6.3).

4.4.5 *Acidity or alkalinity.*—Cut a piece of cloth, weighing about 0.5 gram, from each of 10 sub samples of the lot, and weigh all 10 pieces together. Place the pieces in a covered beaker, containing 150 milliliter (ml.) of distilled water which has been boiling for 10 minutes, and continue boiling for 30 additional minutes. Decant or, if necessary, filter the water, cool it, and titrate it with approximately N/10 acid or alkali, as required, using phenolphthalein as the indicator. Calculate the acidity or alkalinity as percent acetic acid or sodium hydroxide, respectively.

4.5 **Resubmission and retest.**—If the composite sample representing the lot fails to pass the inspection tests, the manufacturer shall have the option of having tests made on each of the primary samples, at no expense

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to the Government. The manufacturer may then replace the defective portions of the lot, corresponding to the primary samples which fail to meet the requirements, and again submit the lot for inspection and acceptance provided that complete replacement of the defective portions has been accomplished to the satisfaction of the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—The silk cloth shall be furnished in rolls, each containing not less than 50 nor more than 400 yards of the cloth. Each roll shall be wound on a substantial cardboard tube that will not distort or break in handling or in shipment. The tube shall be 3 inches in outside diameter, and shall extend a minimum of 2 inches beyond both edges of the cloth. Each roll shall be wrapped with 1 or more layers of heavy manila paper, or of 60-pound basis weight kraft wrapping paper conforming to grade B, No. 2 of Specification UU-P-268. The size of the wrapping paper used shall be such as to permit tucking into the ends of the tube. The wrapper shall be secured by encircling the wrapped roll with gummed paper tape, and shall be sealed by placing a strip of the gummed tape over the entire length of the wrapper seam.

5.2 Packing.—The rolls of cloth, wrapped as specified in 5.1, shall be packed in shipping containers conforming to Specification NN-B-591, NN-B-601, or NN-B-621. Die-cut fiberboard pads or fillers shall be placed between the ends of the rolls and the sides of the container, so as to prevent damage to the roll during shipment.

5.3 Marking.—In addition to any special marking required by the procuring agency each roll of cloth shall be labelled to show the name of the material, class, manufacturer, lot number, contract or purchases order number, and number of yards of cloth in the roll. The shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of cloth required (see 1.2).
(c) Width of the cloth required (see 3.2).
(d) Special markings required (see 5.3).

6.2 Construction of the cloth.—In the construction of silk cloth for ammunition parachutes the thread counts below have been found satisfactory.

6.3 The relationship between the two methods of determining air permeability has been established to be as follows: [1]

$$F = \frac{507.5}{G}$$

where

F = cubic feet of air per square foot per minute, at 0.5 inch - of - water pressure across the fabric (Frazier Permeameter).
G = number of seconds required for 300 centimeter (cc.) of air to flow through the fabric, using an orifice

[1] For a discussion of the established relationship see Textile Research Journal, vol. XVII, No. 4, April 1947, pp. 214-221.

Thread count	Class 1		Class 2	
	Minimum	Maximum	Minimum	Maximum
Warp, threads per inch	130	138	120[1]	135[1]
Filling, threads per inch	100	120	100[1]	125[1]

[1] Double threads, tram twist.

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with an area of 0.10 square inch, and a 5-ounce cylinder (Gurley Densometer).

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